SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105

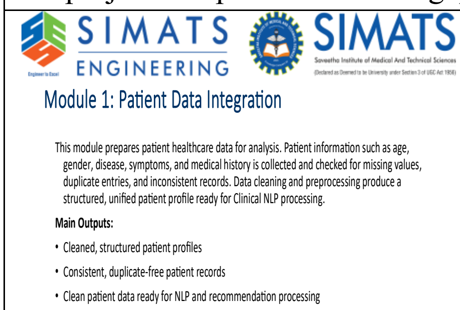
Student Name: Malavathu Saiteja **Reg. No.:** 192424034

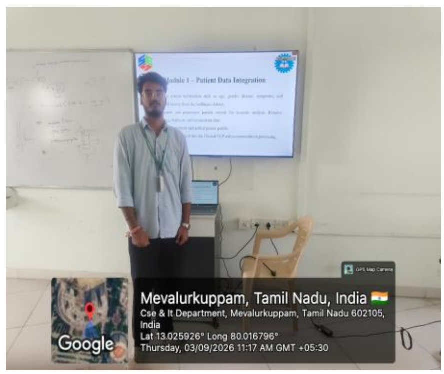
Course Code: DSA0308 **Slot:** D

Course Name: DATA HANDLING AND VISUALIZATION

Course Faculty: Dr. K. Kumaragurabaran

Project Title: Personalized Healthcare Recommendation through Clinical NLP
Module Photographs: (3 photographs – Module Photo, Individual student contribution module work in the project and presentation image)





Project Description:
The **Personalized Healthcare Recommendation through Clinical NLP – Visualization & Analysis Module** is designed to present processed clinical and patient data in an interactive and meaningful way for improved healthcare insights and decision-making. The module utilizes cleaned and structured medical data, including patient history, symptoms, diagnoses, and treatment records, to generate visualizations such as trend charts, distribution plots, correlation heatmaps, and patient risk dashboards. Libraries like Matplotlib, Seaborn, Plotly, and other NLP visualization tools are used to create both static and interactive outputs for better interpretation of clinical data. Time-series analysis is applied to monitor patient health progression, treatment effectiveness, and disease patterns over time. NLP-based visualization techniques highlight key medical terms, symptom frequencies, and relationships between conditions extracted from clinical text. Comparative analysis is performed on different patient groups, diseases, and treatment outcomes to understand effectiveness and risks. The module provides insights such as disease prevalence, patient risk levels, treatment recommendations, and health trend patterns. Interactive dashboards enable users to filter information based on patient demographics, medical conditions, and time periods, making the system efficient, user-friendly, and valuable for personalized healthcare recommendations.

Student Signature **Guide Signature**